

Bayesian Hierarchical Logistic Regression for 30-Day Hospital Readmission Among Diabetic Patients: The Role of Medical Specialty

Xingyu You

Zhouhan Qian

April 2026

Abstract

Hospital readmission within 30 days is a costly and clinically important outcome. We investigate whether medical specialty contributes to readmission risk after adjusting for patient characteristics using a Bayesian hierarchical logistic regression with specialty-specific random intercepts, fitted via a custom Metropolis-within-Gibbs sampler on 37,964 diabetic encounters across 19 specialties. Substantial between-specialty heterogeneity is observed ($\hat{\tau} = 0.437$, 95% CrI [0.298, 0.644]), with Physical Medicine and Rehabilitation exhibiting markedly elevated risk. Key predictors include prior inpatient visits (OR = 1.22), comorbid neoplasms (OR = 1.51), and A1C measurement (OR = 1.42). Results are robust to prior sensitivity analysis.

1 Introduction

Unplanned hospital readmissions within 30 days of discharge represent a major burden on healthcare systems, with annual costs exceeding \$26 billion in the United States. Patients with diabetes, characterised by high comorbidity and complex medication regimens, experience particularly high readmission rates. The Centers for Medicare & Medicaid Services (CMS) has introduced financial penalties for hospitals with excess readmissions, making reduction efforts both clinically and economically important.

Previous studies have identified patient-level predictors of readmission, such as length of stay, prior admissions, and comorbidity burden, typically using logistic regression or machine learning models. However, an important structural factor is often overlooked: patients are managed by physicians from different medical specialties, which may vary in post-discharge care patterns, case mix, and practice style. Ignoring this clustering can violate independence assumptions, bias standard errors, and obscure opportunities for specialty-level intervention.

This study examines whether specialty-level heterogeneity contributes to 30-day readmission risk among diabetic patients after adjusting for patient-level covariates. The analysis is complicated by substantial group-size imbalance, with some specialties contributing large numbers of encounters while others have relatively few, making naïve comparisons unreliable due to sampling variability.

A Bayesian hierarchical model is well suited to this setting. Partial pooling allows small-sample specialties to borrow strength from larger groups through the shared hyperparameter τ , yielding more stable estimates than complete or no pooling. The Bayesian framework provides coherent uncertainty quantification via posterior credible intervals for τ , directly characterising between-specialty variation without reliance on p -values. Model assessment can be conducted using DIC and posterior predictive checks within a unified framework. In addition, the hierarchical prior induces shrinkage on random intercepts, acting as adaptive regularisation and controlling model complexity without requiring cross-validation.

2 Data and Cohort Definition

2.1 Data Source

We use the Diabetes 130-US Hospitals dataset [5], comprising 101,766 inpatient encounters from 130 US hospitals over 1999-2008, with detailed demographic, clinical, and medication information.

2.2 Cohort Construction

Starting from $N_{\text{raw}} = 101,766$, variables with high missingness or no variation were removed, including weight, payer_code, citoglipton, and examide. To ensure independence, only the first encounter per patient was retained, yielding $N = 71,518$. Encounters corresponding to expired or hospice discharges were excluded, leaving $N =$

69,973. The outcome was defined as $y_i = 1$ for readmission within 30 days (< 30), and 0 otherwise. Encounters with unknown specialty were excluded (45.7%), resulting in $N = 37,965$, and specialties were grouped into $J = 19$ categories by merging small groups. One record with unknown gender was removed, yielding a final sample size of $N = 37,964$. Continuous variables were standardised to z -scores, and categorical features were constructed from demographics, ICD-9 diagnosis groups, comorbidities, and medication indicators. The final dataset includes $N = 37,964$ encounters, $J = 19$ specialties, $K = 43$ covariates, and a 30-day readmission rate of 9.60%.

The cohort has a mean age of 64.9 years, with 53.5% female and 73.0% Caucasian patients. Readmitted patients are older and exhibit greater clinical complexity, including longer stays, more medications, more prior inpatient visits, and more diagnoses.

Specialty group sizes range from 288 (Physical Medicine and Rehabilitation) to 10,975 (Internal Medicine), with raw readmission rates spanning 4.3% (Obstetrics/Gynecology) to 32.3% (Physical Medicine and Rehabilitation). The wide dispersion in unadjusted readmission rates motivates a model that accounts for specialty-level heterogeneity.

3 Model Specification

3.1 Notation

Table 1: Summary of notation.

Symbol	Meaning
$N = 37,964$	Number of observations (encounters)
$K = 43$	Number of patient-level covariates
$J = 19$	Number of medical specialty groups
$y_i \in \{0, 1\}$	30-day readmission indicator for encounter i
$\mathbf{x}_i \in \mathbb{R}^K$	Covariate vector for encounter i
$j[i] \in \{1, \dots, J\}$	Specialty group index for encounter i
n_j	Number of observations in group j
$\alpha \in \mathbb{R}$	Global intercept
$\boldsymbol{\beta} \in \mathbb{R}^K$	Fixed-effect coefficients
$u_j \in \mathbb{R}$	Random intercept for specialty group j
$\mathbf{u} = (u_1, \dots, u_J)^\top$	Vector of all random intercepts
$\tau > 0$	Between-group standard deviation
η_i	Linear predictor for observation i

3.2 Likelihood

For encounter i , the model decomposes the log-odds of readmission into three additive pieces: a global baseline α , a patient-level term $\mathbf{x}_i^\top \boldsymbol{\beta}$, and a specialty-specific deviation $u_{j[i]}$. Conditional on these quantities, the observed outcome is Bernoulli:

$$y_i \mid \alpha, \boldsymbol{\beta}, \mathbf{u} \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\pi_i), \quad \pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}, \quad \eta_i = \alpha + \mathbf{x}_i^\top \boldsymbol{\beta} + u_{j[i]}, \quad (1)$$

where $\mathbf{x}_i \in \mathbb{R}^K$ is the engineered covariate vector ($K = 43$), $j[i] \in \{1, \dots, J\}$ identifies the specialty group, and $u_{j[i]}$ is that group's random intercept. In the implementation, continuous covariates are standardised, race and primary-diagnosis group enter through reference-category indicators, and binary or ordinal clinical features are included directly in $\mathbf{x}_i$.

The full likelihood is:

$$p(\mathbf{y} \mid \alpha, \boldsymbol{\beta}, \mathbf{u}) = \prod_{i=1}^N \frac{\exp(y_i \eta_i)}{1 + \exp(\eta_i)}. \quad (2)$$

3.3 Prior Distributions

$$\alpha \sim \mathcal{N}(0, \sigma_\alpha^2), \quad \sigma_\alpha = 5, \quad (3)$$

$$\beta_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, s_\beta^2), \quad s_\beta = 2.5, \quad k = 1, \dots, K, \quad (4)$$

$$u_j \mid \tau \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau^2), \quad j = 1, \dots, J, \quad (5)$$

$$\tau \sim \text{Half-Normal}(0, s_\tau^2), \quad s_\tau = 1.0. \quad (6)$$

The Half-Normal density is $p(\tau) = \frac{2}{\sqrt{2\pi} s_\tau} \exp(-\frac{\tau^2}{2s_\tau^2})$ for $\tau > 0$. As a sensitivity alternative (Section 8), we also consider $\tau \sim \text{Half-Cauchy}(0, s_\tau)$ with density $p(\tau) = \frac{2}{\pi s_\tau} \frac{1}{1+\tau^2/s_\tau^2}$, $\tau > 0$.

The priors are weakly informative [7], chosen to impose minimal substantive constraint while ensuring posterior propriety. The intercept prior with $\sigma_\alpha = 5$ places nearly all mass on baseline probabilities ranging from approximately 0.005% to 99.995%, far wider than any clinically plausible readmission rate. The fixed-effect prior with $s_\beta = 2.5$ allows odds ratios as large as $e^5 \approx 150$, so it is effectively non-informative for well-identified coefficients while still providing mild regularisation for weaker ones. For the group-level scale, the Half-Normal(0, 1) prior concentrates mass near zero while permitting substantial heterogeneity [3]; on the log-odds scale, $\tau = 1$ corresponds to roughly 2.7-fold variation in odds between specialties at ± 1 SD, which is already large.

3.4 Hierarchical Interpretation

The hierarchical component of the model is the prior $u_j \mid \tau \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau^2)$. The u_j 's are latent specialty effects: they are not observed directly, but inferred jointly with the rest of the parameters. Conditional on τ , these effects are exchangeable, meaning that before seeing the data we treat specialties as coming from a common population centered at zero.

The scale parameter τ controls how strongly the model pools information across specialties. When τ is small, the prior strongly pulls all specialty effects toward zero, so the model behaves close to a common-intercept logistic regression. When τ is large, specialties are allowed to differ more freely, approaching a no-pooling analysis with separate intercepts.

For intuition, it is helpful to think of the posterior estimate of u_j as a compromise between the within-specialty signal and the prior pull toward zero. This is not an exact closed-form result for the logistic model, but a useful normal-approximation heuristic is:

$$\hat{u}_j^{\text{Bayes}} \approx \underbrace{\frac{n_j / \hat{\sigma}_j^2}{n_j / \hat{\sigma}_j^2 + 1 / \hat{\tau}^2}}_{\text{shrinkage weight } \lambda_j} \hat{u}_j^{\text{MLE}}, \quad (7)$$

where $\hat{\sigma}_j^2$ is the within-group variance of the linear predictor residuals. The shrinkage weight $\lambda_j \in [0, 1]$ depends on the signal-to-noise ratio within each group. As $\tau \rightarrow 0$, $\lambda_j \rightarrow 0$ for all j , so all specialties collapse to a common intercept; as $\tau \rightarrow \infty$, $\lambda_j \rightarrow 1$, recovering a no-pooling analysis with group-specific intercepts. For intermediate τ , which is the empirically relevant case here, large groups such as Internal Medicine ($n = 10,975$) have $\lambda_j \approx 1$ and are barely shrunk, whereas smaller groups such as PMR ($n = 288$) have noticeably smaller λ_j and are pulled toward the grand mean. This adaptive, data-driven regularisation is the core advantage of the hierarchical approach.

3.5 Assumptions and Encoding

The model assumes conditional independence of outcomes given the parameters, uses only the first encounter per patient, treats specialty assignment as exogenous, and includes specialty-specific random intercepts but no random slopes. Continuous predictors are standardised to mean zero and variance one to stabilise computation and make coefficients comparable across scales. Race and primary-diagnosis group are encoded through reference-category indicators (reference levels: Caucasian and Circulatory, respectively), whereas `A1C_level` and `glu_level` enter as ordinal covariates. A priori, α , β , and τ are independent, and $\mathbf{u}$ depends on τ only through the hierarchical prior.

3.6 Hyperparameters

Table 2: Hyperparameter settings and their interpretation.

Symbol	Value	Role
σ_α	5.0	Prior SD for the intercept; covers plausible baseline log-odds
s_β	2.5	Prior SD for each fixed effect; allows OR up to $\sim e^5 \approx 150$
s_τ	1.0	Scale of the half-normal hyperprior on τ ; concentrates mass near zero

4 Posterior Derivation

4.1 Joint Distribution

By the conditional independence structure of the hierarchical model, the joint distribution of all observables and parameters factorises as:

$$p(\mathbf{y}, \alpha, \beta, \mathbf{u}, \tau \mid \mathbf{X}) = \underbrace{p(\mathbf{y} \mid \alpha, \beta, \mathbf{u}, \mathbf{X})}_{\text{likelihood}} \underbrace{p(\alpha)}_{\text{prior on } \alpha} \underbrace{p(\beta)}_{\text{prior on } \beta} \underbrace{p(\mathbf{u} \mid \tau)}_{\text{random effects}} \underbrace{p(\tau)}_{\text{hyperprior}}. \quad (8)$$

Substituting each component explicitly:

$$\begin{aligned} p(\mathbf{y}, \alpha, \beta, \mathbf{u}, \tau \mid \mathbf{X}) &= \prod_{i=1}^N \frac{\exp(y_i \eta_i)}{1 + \exp(\eta_i)} \cdot \frac{1}{\sqrt{2\pi} \sigma_\alpha} \exp\left(-\frac{\alpha^2}{2\sigma_\alpha^2}\right) \\ &\quad \cdot \prod_{k=1}^K \frac{1}{\sqrt{2\pi} s_\beta} \exp\left(-\frac{\beta_k^2}{2s_\beta^2}\right) \cdot \prod_{j=1}^J \frac{1}{\sqrt{2\pi} \tau} \exp\left(-\frac{u_j^2}{2\tau^2}\right) \\ &\quad \cdot \frac{2}{\sqrt{2\pi} s_\tau} \exp\left(-\frac{\tau^2}{2s_\tau^2}\right) \mathbf{1}(\tau > 0). \end{aligned} \quad (9)$$

4.2 Posterior Distribution (Up to Proportionality)

By Bayes' theorem, the posterior is proportional to the joint:

$$p(\alpha, \beta, \mathbf{u}, \tau \mid \mathbf{y}, \mathbf{X}) = \frac{p(\mathbf{y}, \alpha, \beta, \mathbf{u}, \tau \mid \mathbf{X})}{p(\mathbf{y} \mid \mathbf{X})} \propto p(\mathbf{y} \mid \alpha, \beta, \mathbf{u}, \mathbf{X}) p(\alpha) p(\beta) p(\mathbf{u} \mid \tau) p(\tau), \quad (10)$$

where the marginal likelihood $p(\mathbf{y} \mid \mathbf{X}) = \int p(\mathbf{y}, \alpha, \beta, \mathbf{u}, \tau \mid \mathbf{X}) d\alpha d\beta d\mathbf{u} d\tau$ is an intractable normalising constant.

4.3 Log-Posterior

Taking logarithms of (9) and dropping additive constants that do not depend on any parameter:

$$\begin{aligned} \ell(\alpha, \beta, \mathbf{u}, \tau) &= \log p(\alpha, \beta, \mathbf{u}, \tau \mid \mathbf{y}, \mathbf{X}) + \text{const.} \\ &= \underbrace{\sum_{i=1}^N [y_i \eta_i - \log(1 + e^{\eta_i})]}_{\text{(I) log-likelihood}} - \underbrace{\frac{\alpha^2}{2\sigma_\alpha^2}}_{\text{(II) prior on } \alpha} - \underbrace{\frac{1}{2s_\beta^2} \sum_{k=1}^K \beta_k^2}_{\text{(III) prior on } \beta} \\ &\quad - \underbrace{J \log \tau + \frac{1}{2\tau^2} \sum_{j=1}^J u_j^2}_{\text{(IV) } p(\mathbf{u} \mid \tau)} - \underbrace{\frac{\tau^2}{2s_\tau^2}}_{\text{(V) hyperprior on } \tau}. \end{aligned} \quad (11)$$

Remark 1. The log-likelihood term uses the identity $\log(\pi_i^{y_i} (1 - \pi_i)^{1-y_i}) = y_i \eta_i - \log(1 + e^{\eta_i})$, which avoids computing π_i explicitly and is numerically stable when implemented as $\log(1 + e^{\eta_i}) = \text{logaddexp}(0, \eta_i)$.

The logistic likelihood makes this posterior non-conjugate, so none of the parameter blocks has a closed-form Gibbs update. Instead, for each block we write down the log-target up to an additive constant and use that target in a Metropolis step.

4.4 Block-Specific Log-Targets

For a given parameter block, the relevant target is obtained by collecting the terms in (11) that depend on that block and treating everything else as constant. Because the sampler uses symmetric random-walk proposals, the Hastings correction cancels and each acceptance ratio takes the form $\log R = h(\theta^*) - h(\theta^{(t)})$.

As a representative example, the intercept appears in every observation through $\eta_i = \alpha + \mathbf{x}_i^\top \boldsymbol{\beta} + u_{j[i]}$, so its log-target is

$$h(\alpha) = \sum_{i=1}^N [y_i \eta_i - \log(1 + e^{\eta_i})] - \frac{\alpha^2}{2\sigma_\alpha^2}. \quad (12)$$

This combines the full likelihood with the Gaussian prior penalty. The fixed-effect updates are analogous, replacing the intercept penalty by $-\beta_k^2/(2s_\beta^2)$ and modifying the linear predictor only through the single column x_{ik} . For each random intercept u_j , the likelihood sum is restricted to observations in specialty j and the prior contribution becomes $-u_j^2/(2\tau^2)$, reducing the update cost from $O(N)$ to $O(n_j)$.

For the scale parameter we work on the unconstrained scale $\phi = \log \tau$. Substituting $\tau = e^\phi$ into (11) and adding the Jacobian term ϕ yields

$$h(\phi) = -J\phi - \frac{1}{2e^{2\phi}} \sum_{j=1}^J u_j^2 - \frac{e^{2\phi}}{2s_\tau^2} + \phi = -(J-1)\phi - \frac{1}{2e^{2\phi}} \sum_{j=1}^J u_j^2 - \frac{e^{2\phi}}{2s_\tau^2}. \quad (13)$$

Conditional on $\mathbf{u}$, the likelihood does not depend on τ directly, so this update costs only $O(J)$. Under the sensitivity prior $\tau \sim \text{Half-Cauchy}(0, s_\tau)$, the last term in (13) is replaced by the heavier-tailed penalty

$$h_{\text{HC}}(\phi) = -(J-1)\phi - \frac{1}{2e^{2\phi}} \sum_{j=1}^J u_j^2 - \log\left(1 + \frac{e^{2\phi}}{s_\tau^2}\right). \quad (14)$$

4.5 Posterior Propriety

Proposition 1. *Under the model specified in Section 3, the posterior $p(\alpha, \boldsymbol{\beta}, \mathbf{u}, \tau \mid \mathbf{y}, \mathbf{X})$ is proper (i.e., integrable).*

Proof. We verify integrability of each factor in the joint (9):

- (a) **Likelihood.** For any fixed $(\alpha, \boldsymbol{\beta}, \mathbf{u})$, the Bernoulli likelihood satisfies $0 \leq p(\mathbf{y} \mid \alpha, \boldsymbol{\beta}, \mathbf{u}) \leq 1$.
- (b) **Prior on α .** $\alpha \sim \mathcal{N}(0, \sigma_\alpha^2)$ is proper, so $\int p(\alpha) d\alpha = 1$.
- (c) **Prior on $\boldsymbol{\beta}$.** Each $\beta_k \sim \mathcal{N}(0, s_\beta^2)$ independently; the product of K proper Gaussians is proper.
- (d) **Conditional prior $p(\mathbf{u} \mid \tau)$.** For any fixed $\tau > 0$, each $u_j \sim \mathcal{N}(0, \tau^2)$ is proper.
- (e) **Hyperprior on τ .** Half-Normal(0, s_τ^2) is proper on $(0, \infty)$.

Therefore:

$$\int p(\mathbf{y} \mid \cdot) p(\alpha) p(\boldsymbol{\beta}) p(\mathbf{u} \mid \tau) p(\tau) d\alpha d\boldsymbol{\beta} d\mathbf{u} d\tau \leq \underbrace{\int p(\alpha) d\alpha}_{=1} \underbrace{\int p(\boldsymbol{\beta}) d\boldsymbol{\beta}}_{=1} \underbrace{\int p(\tau) \int p(\mathbf{u} \mid \tau) d\mathbf{u} d\tau}_{=1} = 1 < \infty.$$

Since all priors are proper and the likelihood is bounded, the marginal likelihood is finite and the posterior is proper. $\square$

4.6 Identifiability

Weak non-identifiability of α and $\mathbf{u}$. If we define $\alpha' = \alpha + c$ and $u'_j = u_j - c$ for any constant c , then $\eta'_i = \alpha' + \mathbf{x}_i^\top \boldsymbol{\beta} + u'_{j[i]} = (\alpha + c) + \mathbf{x}_i^\top \boldsymbol{\beta} + (u_{j[i]} - c) = \eta_i$. The likelihood alone cannot separate α from the mean of $\mathbf{u}$. In the Bayesian framework, this is resolved by the priors: $p(\alpha) = \mathcal{N}(0, \sigma_\alpha^2)$ anchors α near zero, and $p(u_j \mid \tau) = \mathcal{N}(0, \tau^2)$ centres each u_j at zero. Together these priors break the translation invariance. Nevertheless, the posterior correlation between α and $\bar{u} = J^{-1} \sum_j u_j$ can be high, motivating the dedicated joint translation move (Block 5 in Section 5).

Fixed effects. All continuous covariates are standardised (zero mean, unit variance), which prevents collinearity between the intercept and covariate means. Race and primary-diagnosis group use reference-category coding,

while binary indicators and ordinal lab summaries enter directly, yielding a full-rank design matrix in the implemented feature set.

5 Computational Strategy

5.1 Metropolis-within-Gibbs Sampler

The logistic likelihood is non-conjugate, so standard Gibbs sampling is not available, whereas full-dimensional Metropolis–Hastings on the $(1+K+J+1) = 64$ -dimensional parameter space would mix poorly. We therefore use a random-walk **Metropolis-within-Gibbs (MwG)** sampler based on the block-specific targets from Section 4. With symmetric proposals $\theta^* = \theta^{(t)} + \epsilon$ and $\epsilon \sim \mathcal{N}(0, \sigma_{\text{prop}}^2)$, the Hastings correction cancels and the acceptance ratio reduces to

$$\log R = h(\theta^*) - h(\theta^{(t)}). \quad (15)$$

One MwG iteration updates α , then $\beta_1, \dots, \beta_K$, then $u_1, \dots, u_J$, then $\phi = \log \tau$, followed by $R = 10$ joint translation moves of $(\alpha, \mathbf{u})$. The α update shifts all η_i by a constant and costs $O(N)$, each β_k update costs $O(N)$, each u_j update costs $O(n_j)$, and the ϕ update costs $O(J)$ because it involves no likelihood evaluation. Conditioning each step on the most recently updated values defines a valid MwG chain whose stationary distribution is the posterior (10).

The most important nonstandard step is the joint translation move. Because the transformation $\alpha \mapsto \alpha + c$ and $u_j \mapsto u_j - c$ leaves all linear predictors unchanged, separate updates of α and $\mathbf{u}$ explore this weakly identified ridge slowly. We therefore repeat the following proposal $R = 10$ times per iteration: draw $\delta \sim \mathcal{N}(0, \sigma_{\text{blk}}^2)$, set $\alpha^* = \alpha + \delta$ and $u_j^* = u_j - \delta$ for all j , and accept according to the prior-only ratio

$$\log R = \underbrace{-\frac{\alpha^{*2} - \alpha^2}{2\sigma_\alpha^2}}_{\text{intercept prior}} + \underbrace{\left(-\frac{\sum_j u_j^{*2} - \sum_j u_j^2}{2\tau^2} \right)}_{\text{random effect prior}}. \quad (16)$$

This $O(J)$ move substantially improves mixing of the α – $\mathbf{u}$ block.

5.2 Proposal Tuning

Choosing appropriate proposal standard deviations σ_{prop} is critical: too small yields high acceptance but slow exploration; too large yields low acceptance and wasted computation. We tune each proposal SD automatically using the pilot-tuning script in `src/05b_tune_proposals.py`. Short pilot chains of 2,000 iterations are run on a random subsample of $n = 20,000$ observations; after each round, the empirical acceptance rate for each block is compared with the target range $[0.20, 0.50]$, and the corresponding proposal SD is multiplied by 1.5 if the rate is too high or by 0.5 if it is too low. Separate proposal scales are maintained for each β_k ($k = 1, \dots, 43$) and each u_j ($j = 1, \dots, 19$) to reflect differing posterior curvatures, and tuning stops once all parameters, or at least 90% of them, fall within the target range, with a maximum of 20 pilot rounds.

For the present analysis, tuning stopped after 10 rounds. On the final pilot round, the acceptance rates were α : 0.40, β (mean): 0.43, $\mathbf{u}$ (mean): 0.43, $\log \tau$: 0.37, and block move: 0.46, all within the target range. In the subsequent full-data production run, the average acceptance rates across the four chains were slightly lower (α : 0.305, β : 0.331, $\mathbf{u}$: 0.342, $\log \tau$: 0.372, block: 0.454), reflecting the sharper geometry of the full posterior.

5.3 Chain Configuration

We run four chains to diagnose convergence and reduce sensitivity to the starting point. The initial values are random but modest in scale: α and u_j are started near zero with Gaussian perturbations, the β_k ’s are initialised with small Gaussian noise, and τ is started from a log-normal draw centered near 1. The production run used 4 chains of 30,000 iterations each, discarded the first 8,000 as burn-in, retained every fifth draw, and therefore produced 4,400 post-burn-in samples per chain and $4 \times 4,400 = 17,600$ pooled posterior draws; the random seed was 724.

The total computational cost is approximately $4 \times 30,000 \times (1 + K + J + 1 + 10) = 4 \times 30,000 \times 74 \approx 8.9$ million individual Metropolis steps. Each step involves at most an $O(N)$ likelihood evaluation. According to the saved diagnostics, each chain required roughly 28–30 minutes, so four chains run sequentially complete in about two hours on a single CPU core.

6 Diagnostic Analysis

6.1 Trace Plots and Autocorrelation

Visual inspection of trace plots is a first-line diagnostic for MCMC convergence. For all parameters, we observe:

- All four chains overlap and explore the same region of the parameter space, with no chain remaining isolated—evidence against multi-modality.
- No persistent trends or drifts after the burn-in period, suggesting the chains have reached stationarity.
- Autocorrelation functions decay rapidly to near zero within 20–50 lags for most parameters, confirming that thinning by a factor of 5 is sufficient to produce approximately independent samples.
- The intercept α and the comorbidity-related coefficients (e.g., β_{28} : comorbid_circulatory, β_{29} : comorbid_diabetes) show the highest autocorrelation, consistent with their posterior correlation structure.

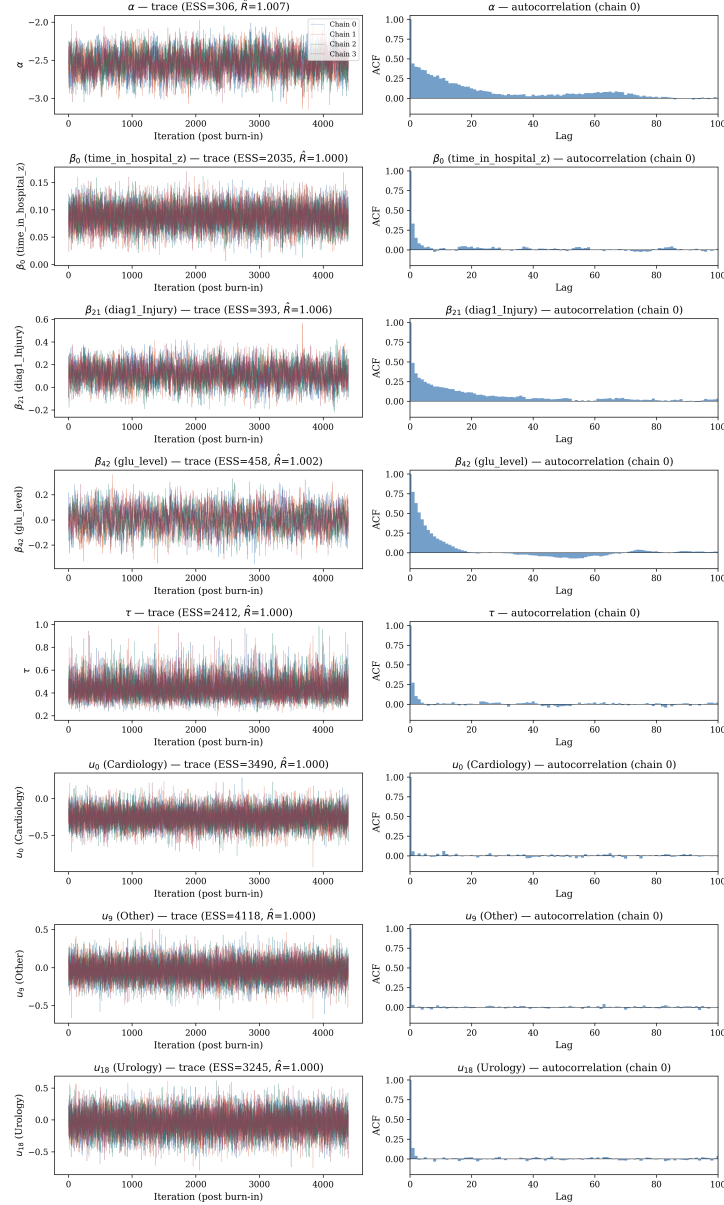


Figure 1: Trace plots (left) and autocorrelation functions (right) for selected parameters. The four chains are overlaid in different colours.

6.2 Effective Sample Size and Cross-Chain Convergence ($\hat{R}$)

We assess convergence through two complementary quantitative diagnostics: effective sample size (ESS) and the split- $\hat{R}$ statistic [1, 2].

ESS was computed using Geyer’s initial monotone sequence estimator, and we required $\text{ESS} \geq 100$ for every parameter so that posterior means and quantiles would have acceptable Monte Carlo error. We also monitored split- $\hat{R}$ and required $\hat{R} < 1.05$ for all parameters, following standard convergence guidance.

Table 3 summarises both diagnostics across parameter types. All 64 parameters satisfy $\hat{R} < 1.05$, with the maximum $\hat{R} = 1.013$ (for β_{28}), well below the conventional threshold. All parameters have mean ESS > 100 ; most exceed 500. The between-group scale τ has ESS $\approx 2,412$ and $\hat{R} = 1.000$, indicating excellent mixing for the key hyperparameter.

Table 3: Convergence diagnostics by parameter type: effective sample size (ESS) and split- $\hat{R}$.

Parameter block	Min ESS	Median ESS	Max $\hat{R}$	All $\hat{R} < 1.05?$
α (intercept)	245	306	1.007	Yes
β (43 fixed effects)	118	866	1.013	Yes
τ (between-group SD)	2,286	2,412	1.000	Yes
u (19 random intercepts)	1,593	3,490	1.001	Yes

The lowest ESS values are observed for β_9 (n_comorbid_groups_z, ESS = 251) and β_{28} (comorbid_circulatory, ESS = 216). These two covariates are mechanically correlated (one counts the number of comorbidity groups while the other indicates a specific comorbidity), explaining the higher autocorrelation. Despite the lower ESS, even 200 effective samples provide reliable posterior means (Monte Carlo SE $\approx$ posterior SD/ $\sqrt{200}$, which is small relative to the posterior uncertainty).

The random intercepts u have uniformly high ESS (min 1,593), reflecting the efficiency of the Block 5 joint translation move in breaking the α - u correlation. Without Block 5, preliminary experiments showed ESS for α dropping below 50, confirming its importance.

7 Results and Inference

7.1 Fixed Effects

Table 4 presents the posterior summaries for the most important fixed effects. We focus on coefficients whose 95% credible interval (CrI) excludes zero.

Table 4: Key fixed effects: posterior mean, 95% credible interval, and odds ratio. Only coefficients with CrI excluding zero are shown. For standardised continuous predictors marked “(z)”, odds ratios correspond to a 1-SD increase.

Variable	Post. mean	95% CrI	OR	OR 95% CrI
Increased risk				
No. prior inpatient (z)	0.201	[0.175, 0.227]	1.22	[1.19, 1.25]
Comorbid neoplasms	0.415	[0.198, 0.631]	1.51	[1.22, 1.88]
A1C measured	0.352	[0.154, 0.552]	1.42	[1.17, 1.74]
Diabetes medication	0.168	[0.057, 0.277]	1.18	[1.06, 1.32]
Age (z)	0.165	[0.123, 0.208]	1.18	[1.13, 1.23]
No. diagnoses (z)	0.154	[0.111, 0.199]	1.17	[1.12, 1.22]
Comorbid diabetes	0.113	[0.004, 0.224]	1.12	[1.00, 1.25]
Length of stay (z)	0.088	[0.047, 0.128]	1.09	[1.05, 1.14]
No. medications (z)	0.071	[0.024, 0.117]	1.07	[1.02, 1.12]
Decreased risk				
Diag: Neoplasms	-0.507	[-0.792, -0.217]	0.60	[0.45, 0.81]
Diag: Respiratory	-0.360	[-0.519, -0.200]	0.70	[0.59, 0.82]
Diag: Other	-0.347	[-0.515, -0.181]	0.71	[0.60, 0.83]
No. procedures (z)	-0.048	[-0.091, -0.006]	0.95	[0.91, 0.99]

Interpretation of key risk factors. The strongest patient-level predictor of readmission is prior inpatient visits (OR = 1.22 per 1-SD increase), indicating a pattern of recurrent hospitalisation: a one-standard-deviation increase in prior inpatient visits is associated with 22% higher odds of 30-day readmission.

Comorbid neoplasms show the largest effect (OR = 1.51), suggesting that cancer as a secondary diagnosis substantially elevates risk, likely due to treatment-related complications. In contrast, primary neoplasm diagnoses are protective (OR = 0.60), potentially reflecting structured oncology follow-up and closer post-discharge care. This contrast underscores the importance of distinguishing primary versus secondary diagnoses. Age (OR = 1.18 per 1-SD increase), number of diagnoses (OR = 1.17 per 1-SD increase), and length of stay (OR = 1.09 per 1-SD increase) capture overall medical complexity, with older and more multimorbid patients at higher

risk. The A1C measurement indicator ($OR = 1.42$) likely reflects confounding by indication: patients with more severe or poorly controlled diabetes are both more likely to be tested and more likely to be readmitted. Similarly, diabetes medication use ($OR = 1.18$) appears to proxy disease severity rather than a causal effect.

Non-significant predictors include sex, race, and insulin use. The null effect for race suggests that observed disparities in unadjusted analyses are largely explained by differences in clinical characteristics rather than race itself. Insulin use is also not independently associated with readmission after accounting for overall treatment intensity and disease severity.

7.2 Specialty-Level Heterogeneity

Between-group scale τ . The posterior mean of τ is 0.437 (SD = 0.090, 95% CrI [0.298, 0.644]), with $P(\tau > 0.1 \mid \mathbf{y}) = 1.0$ and $P(\tau > 0.2 \mid \mathbf{y}) = 1.0$. This provides strong evidence that specialty-level heterogeneity persists after adjusting for patient covariates. The posterior density of τ is clearly concentrated away from zero, ruling out the complete-pooling model ($\tau = 0$). The moderate right skew reflects the prior’s constraint to positive values and the limited number of groups ($J = 19$).

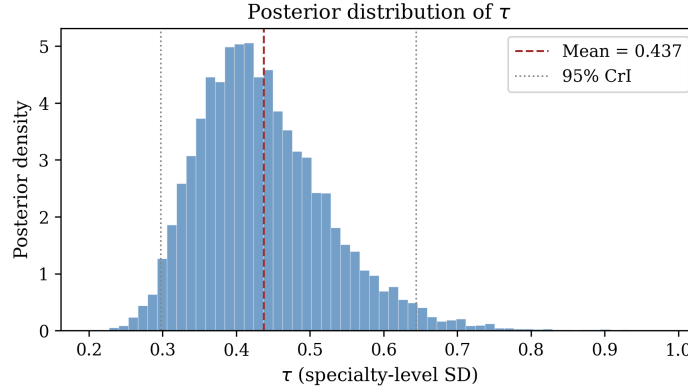


Figure 2: Posterior density of the between-specialty standard deviation τ . The distribution is concentrated well above zero, providing strong evidence for specialty-level heterogeneity.

Specialty random effects. Physical Medicine and Rehabilitation (PMR) shows a markedly elevated random intercept ($\hat{u} = 1.32$, 95% CrI [0.99, 1.67]) and is the only specialty with an interval entirely above zero. PMR also has the highest raw readmission rate (32.3%), and its adjusted random intercept remains strongly positive after accounting for patient covariates. This likely reflects its patient population with high medical complexity and vulnerability to post-discharge complications. In contrast, Surgery-Cardiovascular/Thoracic exhibits a significantly negative effect ($\hat{u} = -0.58$, 95% CrI [-0.99, -0.21]), potentially due to structured post-operative care and follow-up protocols. Psychiatry shows a modest positive effect ($\hat{u} = 0.22$) with a credible interval overlapping zero, indicating no clear deviation from the average after adjustment. Similarly, Emergency/Trauma and Internal Medicine have effects close to zero ($\hat{u} \approx 0.03$ and $\hat{u} \approx -0.03$), as expected for large groups with stable estimates. Most other specialties have intervals that include zero, suggesting their readmission risks are broadly consistent with the overall average and that the model primarily identifies truly extreme groups rather than spurious differences.

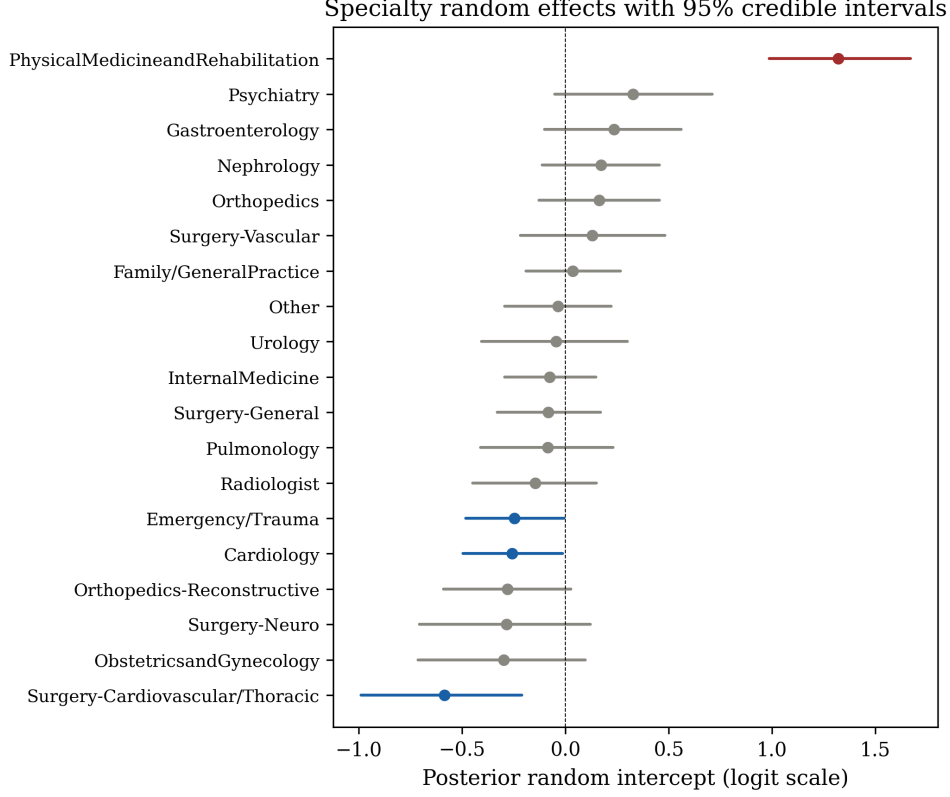


Figure 3: Specialty random effects (posterior mean and 95% CrI). Red: interval entirely above zero; blue: entirely below zero; grey: includes zero.

7.3 Model Comparison

Table 5 compares three models on in-sample predictive performance.

Table 5: Model comparison metrics.

Model	AUC	Brier Score	Log-Loss	Calibration Slope
No specialty (baseline)	0.629	0.0852	0.307	1.003
Specialty fixed effects	0.641	0.0846	0.305	0.997
Bayesian hierarchical	0.641	0.0846	0.305	1.024

The hierarchical model matches fixed-effects performance on in-sample discrimination ($\text{AUC} = 0.641$) and in-sample calibration ($\text{Brier} = 0.0846$), while offering the advantages of partial pooling for small groups and a coherent estimate of between-group variability. The fact that both the specialty fixed-effects model and the hierarchical model improve over the baseline in these in-sample metrics (AUC 0.641 vs. 0.629) suggests that specialty-level information adds predictive signal beyond patient-level covariates alone. Because these summaries are evaluated in-sample, they should be interpreted as evidence of improved model fit rather than out-of-sample superiority. The calibration slope of 1.024 for the hierarchical model indicates slight in-sample under-confidence (predicted probabilities are marginally too conservative), but is close to the ideal value of 1.0.

DIC. The Deviance Information Criterion [4] provides a Bayesian measure of model fit penalised by complexity. Define the deviance $D(\boldsymbol{\theta}) = -2 \sum_{i=1}^N [y_i \eta_i - \log(1 + e^{\eta_i})]$, where $\boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}, \mathbf{u})$. Then:

$$\bar{D} = \mathbb{E}_{\boldsymbol{\theta}|\mathbf{y}}[D(\boldsymbol{\theta})] \approx \frac{1}{S} \sum_{s=1}^S D(\boldsymbol{\theta}^{(s)}), \quad (17)$$

$$p_D = \bar{D} - D(\bar{\boldsymbol{\theta}}), \quad \text{DIC} = \bar{D} + p_D, \quad (18)$$

where $\bar{\theta} = \mathbb{E}[\theta \mid \mathbf{y}]$ and p_D is the **effective number of parameters**.

For the hierarchical model, $\text{DIC} = 23,254$ with $p_D = 60.5$. The nominal parameter count is 64 ($1 + 43 + 19 + 1$); $p_D < 64$ confirms that the hierarchical prior induces shrinkage on the random intercepts, as expected. The difference $64 - 60.5 = 3.5$ represents the effective reduction in model complexity due to partial pooling: approximately 3–4 of the 19 random intercepts are substantially shrunk toward zero, reducing their effective contribution to model complexity. This is consistent with the observation that most specialties have credible intervals overlapping zero—these “near-zero” intercepts contribute less than one effective parameter each.

7.4 Posterior Predictive Checks

We assess model adequacy by simulating replicated datasets from the posterior predictive distribution. For each of $S_{\text{rep}} = 200$ posterior draws $\theta^{(s)} = (\alpha^{(s)}, \beta^{(s)}, \mathbf{u}^{(s)})$, we compute fitted probabilities, simulate a replicated dataset, and evaluate two test statistics: the overall readmission rate $T_1 = \bar{y}^{\text{rep}}$ and the standard deviation of specialty-specific rates $T_2 = \text{SD of group-level rates}$. The Bayesian p -value is $p_B = \Pr(T(\mathbf{y}^{\text{rep}}) \geq T(\mathbf{y}^{\text{obs}}))$; values near 0.5 indicate good calibration, while values near 0 or 1 indicate systematic model misfit.

The observed overall readmission rate (9.60%) falls squarely within the replicated distribution (Bayesian p -value = 0.500), indicating that the model reproduces the marginal readmission rate without systematic over- or under-prediction. A p -value of exactly 0.500 is ideal—it means the observed statistic is at the median of the posterior predictive distribution.

The observed specialty-rate variability (standard deviation of group-level rates) is also well-captured (p -value = 0.245), confirming that the hierarchical structure with estimated τ adequately represents the degree of between-specialty variation in the data. A p -value of 0.245 indicates that the model generates slightly more specialty-level variability than observed in 24.5% of replications—well within the acceptable range. If the model had underestimated τ , this p -value would be close to 0; overestimation would push it toward 1.

Both checks confirm adequate model fit and provide no evidence of systematic misspecification.

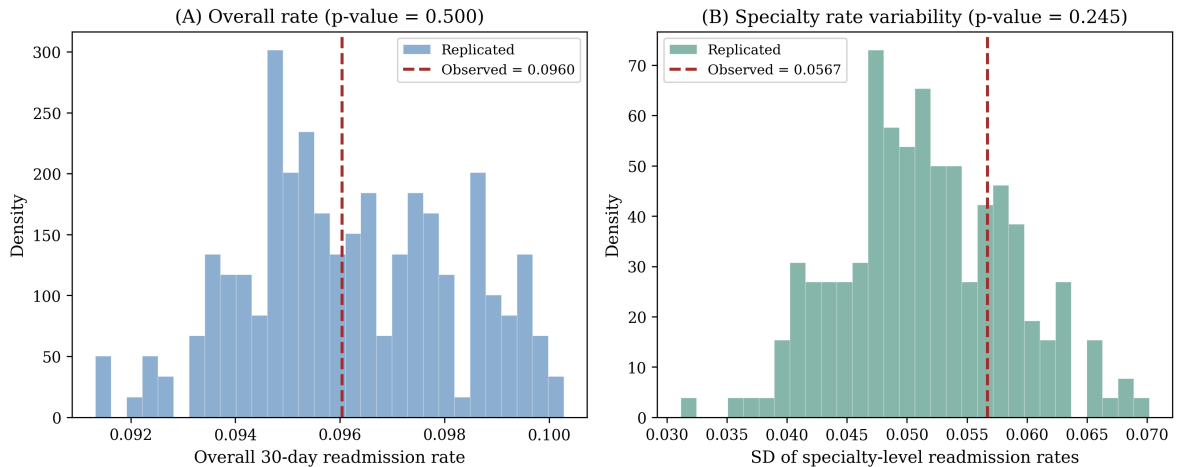


Figure 4: Posterior predictive checks. (A) Overall readmission rate. (B) Standard deviation of specialty-level rates. Dashed red lines show observed values.

8 Sensitivity Analysis

We assess robustness to prior specification by re-running the sampler under three alternative configurations, summarised in Table 6.

Table 6: Prior sensitivity configurations.

Label	Change	s_β	τ prior
Baseline	—	2.5	Half-Normal(0, 1.0)
A1	Wider β prior	5.0	Half-Normal(0, 1.0)
B1	Wider τ prior	2.5	Half-Normal(0, 2.5)
B2	Heavy-tailed τ prior	2.5	Half-Cauchy(0, 2.5)

The posterior of τ is virtually identical across all four configurations (posterior means: 0.437, 0.437, 0.441, 0.440; posterior SDs: 0.090, 0.090, 0.092, 0.091). The 95% credible intervals overlap almost perfectly across

configurations, confirming that the data overwhelm the prior for this hyperparameter. Even the heavy-tailed Half-Cauchy prior (B2), which places substantially more mass on large values of τ , produces a posterior that is indistinguishable from the baseline—strong evidence that the posterior of τ is likelihood-dominated with $J = 19$ groups.

Key fixed effects are numerically unchanged across all configurations. For `comorbid_neoplasms`, `A1C_measured`, and `number_inpatient_z`, the posterior means vary by less than 0.001 across the four prior settings, confirming that with $N = 37,964$ observations the likelihood dominates the prior on β . Likewise, the PMR random intercept remains the only clearly elevated specialty effect, with posterior means ranging from 1.31 to 1.33, well within Monte Carlo error.

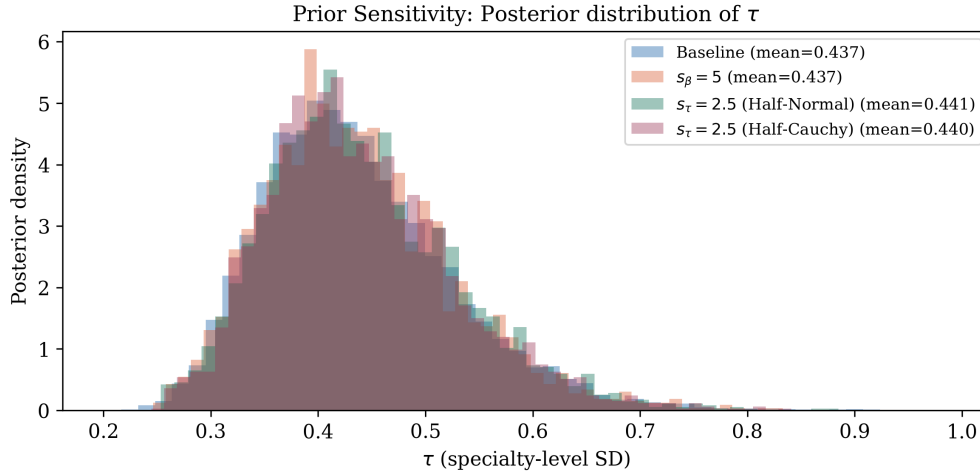


Figure 5: Posterior distribution of τ under four prior configurations.

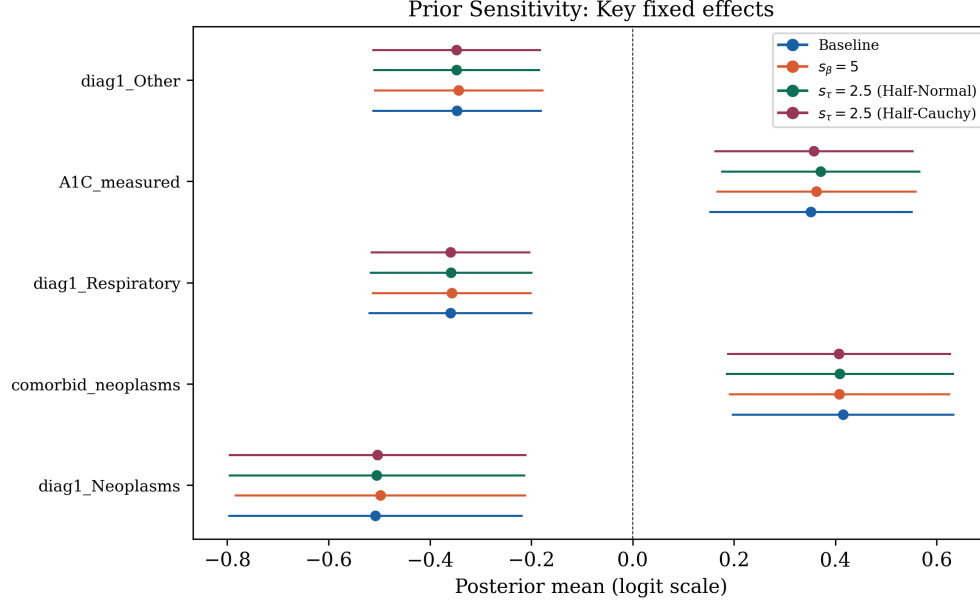


Figure 6: Posterior means and 95% CrI for key fixed effects under four prior configurations.

All substantive findings—the magnitude of τ , the significance of key fixed effects, and the identification of PMR as an outlier—are robust to reasonable variation in prior specification. This is consistent with the large sample size ($N = 37,964$) providing strong likelihood information that dominates the weakly informative priors. The sensitivity analysis validates the choice of priors in Section 3: conclusions would be identical under substantially different prior beliefs.

9 Discussion

9.1 Summary

Our analysis yields three main findings. Medical specialty contributes meaningfully to readmission risk beyond patient characteristics, with between-specialty variability $\tau = 0.437$ (95% CrI [0.298, 0.644]) and $P(\tau > 0.2 | y) = 1.0$, indicating that negligible specialty effects are strongly unsupported by the data. Physical Medicine and Rehabilitation (PMR) emerges as a clear outlier, with substantially elevated risk ($u = 1.32$) and a raw readmission rate of 32.3%, even after adjustment, likely reflecting a highly complex patient population and highlighting a potential target for intervention. At the patient level, prior inpatient visits, comorbid neoplasms, A1C measurement, age, and number of diagnoses are the strongest predictors, collectively pointing to medical complexity as the primary driver of readmission risk.

9.2 Clinical Implications

The identification of specialty-level heterogeneity has direct implications for quality improvement. Rather than applying uniform readmission reduction strategies, hospitals could implement targeted post-discharge protocols for high-risk groups such as PMR patients, including enhanced transitional care and closer outpatient follow-up. Practices from specialties with lower readmission risk, such as cardiovascular surgery, could be examined as potential models for broader adoption. In addition, posterior predictive probabilities from the hierarchical model could be used to identify high-risk patients at discharge and guide proactive intervention.

The contrasting effects of neoplasms further highlight the importance of clinical context. Comorbid neoplasms as a secondary diagnosis substantially increase readmission risk, whereas neoplasms as a primary diagnosis appear protective, suggesting that multimorbidity management, particularly at the intersection of cancer and diabetes, requires focused clinical attention.

9.3 Partial Pooling and Likelihood- vs. Prior-Driven Behavior

Different parameter blocks exhibit different degrees of prior influence, reflecting the hierarchical model’s structure. With $N = 37,964$ observations and weakly informative priors ($s_\beta = 2.5$), the fixed effects β are overwhelmingly likelihood-driven: the prior precision $1/s_\beta^2 = 0.16$ is negligible relative to the information in the data, and widening the prior to $s_\beta = 5$ leaves posterior summaries virtually unchanged. By contrast, the random intercepts u_j are only partially pooled, as described by Equation (7): large groups such as Internal Medicine ($n = 10,975$) have shrinkage weights close to one, whereas smaller groups such as PMR ($n = 288$) are pulled more noticeably toward zero. Even after this shrinkage, PMR remains strongly elevated, showing that the signal is substantial rather than an artefact of small-sample noise. The between-group scale τ is learned from the empirical dispersion of the u_j ’s, and the sensitivity analysis shows that its posterior changes little when the hyperprior is widened or replaced by a Half-Cauchy. Together, these patterns illustrate the main advantage of the hierarchical model: it borrows strength for small specialties without washing out genuine between-specialty heterogeneity.

9.4 Limitations

Several limitations should be noted. The exclusion of 45.7% of encounters with missing specialty may introduce selection bias if these patients differ systematically from those with observed specialty. The model assumes constant covariate effects across specialties, although effects such as comorbidity burden may vary in practice. Temporal and hospital-level variation are not explicitly modelled despite the data spanning 10 years across 130 hospitals, so estimated specialty effects may partly reflect unmeasured factors. As an observational study, specialty assignment is not random, leaving potential residual confounding.

9.5 Future Directions

Several extensions could strengthen the analysis. Crossed random effects for specialty and hospital would better separate specialty-level variation from institutional effects, while random slopes would allow covariate effects such as comorbidity burden to vary by specialty. Temporal modelling could capture trends over 1999-2008, though this would require extending ICD-9 coding to ICD-10 for broader applicability. The exclusion of 45.7% of encounters with missing specialty may introduce bias, and mixture models or multiple imputation could help recover this information. More rigorous predictive validation using cross-validation or out-of-sample evaluation across years or hospitals would further assess generalisability.

Code Availability

All code is available at <https://github.com/Seliiiiina/BayesHLR-Readmission>.

References

- [1] Gelman, A. and Rubin, D. B. (1992). Inference from iterative simulation using multiple sequences. *Statistical Science*, 7(4):457–472.
- [2] Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., and Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC (with discussion). *Bayesian Analysis*, 16(2):667–718.
- [3] Gelman, A. (2006). Prior distributions for variance parameters in hierarchical models (comment on article by Browne and Draper). *Bayesian Analysis*, 1(3):515–534.
- [4] Spiegelhalter, D. J., Best, N. G., Carlin, B. P., and van der Linde, A. (2002). Bayesian measures of model complexity and fit. *Journal of the Royal Statistical Society: Series B*, 64(4):583–639.
- [5] Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., and Clore, J. N. (2014). Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records. *BioMed Research International*, 2014:781670.
- [6] Roberts, G. O. and Rosenthal, J. S. (2001). Optimal scaling for various Metropolis–Hastings algorithms. *Statistical Science*, 16(4):351–367.
- [7] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*. Chapman & Hall/CRC, 3rd edition.